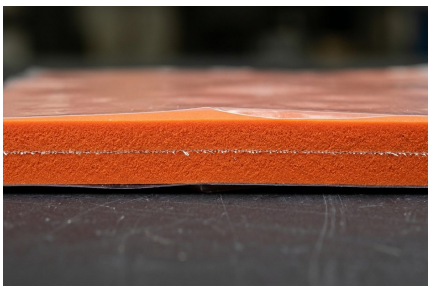
CORE SERIES (Core-300 / Core-350D)

InsulTherm Core™

5-Layer Tropical Anti-Bleed Vertical Conduction TIM Gap Pad (3.0 W/m·K)

InsulTherm Core™ is a 5-layer, fiberglass-reinforced viscoelastic thermal interface gap filler pad engineered specifically for 50°C Indian tropical ambient extremes, rapid EV battery R&D homologation, and industrial BMS power stages. Compounded with 72% wt silane-passivated spherical alumina (Al2O3) inside a crosslinked PDMS matrix around a central 0.05 mm woven E-Glass anti-tear scrim, Core™ eliminates siloxane oil bleed, pump-out, and bolt-hole tearing on rough cast cold plates.



Macro Cross-Section / Studio Hardware Inspection

k = 3.0 W/m·K	Rth 0.22–0.28 K·cm²/W	40 Shore 00	50°C Anti-Bleed	UL 94 V-0

1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Addition-cured vinyl-functionalized polydimethylsiloxane (PDMS) elastomer filled at 72% wt with surface silane-coupled spherical aluminum oxide (Al2O3) ceramics. Silane passivation chemically anchors filler particles to the polymer backbone, preventing free-oil separation during 50°C–150°C cycling, while a neutral-axis 0.05 mm E-Glass scrim prevents stretch distortion and burr puncture.

Layer 1 (0.050 mm)	Top Extended-Tab Clear PET Release Liner (Glove-Friendly Pull-Tab)
Layer 2 (Upper TIM)	Viscoelastic PDMS Silicone + 72% wt Silane-Passivated Al2O3 (3.0 W/m·K)
Layer 3 (0.050 mm)	Center-Embedded Woven E-Glass Fiberglass Anti-Tear & Burr Scrim
Layer 4 (Lower TIM)	Compliant Natural-Tack PDMS + 72% wt Spherical Al2O3 Matrix
Layer 5 (0.075 mm)	Differential-Release Bottom PET Carrier Film (CNC Kiss-Cut Base)

2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

PROPERTY / PARAMETER	TYPICAL ENGINEERING VALUE	TEST METHOD / STANDARD
Bulk Thermal Conductivity	3.0 W/m·K	ASTM D5470-17 Steady-State
Thermal Impedance @ 1mm, 20 psi	0.22 – 0.28 K·cm²/W (0.14 – 0.18°C·in²/W)	ASTM D5470-17 Calibration Rig
Hardness (Bulk Conformability)	40 Shore 00 (Low PCB & cell mounting stress)	ASTM D2240 (3-sec delay)
Tropical Anti-Bleed Stability	Zero oil bleed / pump-out (1,000h @ 50°C–150°C)	InsulTherm Tropical Aging
Dielectric Breakdown Voltage	> 10 kV/mm (Core-300) / > 14 kV/mm (Core-350D)	ASTM D149
Volume Resistivity	> 1.0 × 10^13 Ω·cm	ASTM D257
Flammability Rating	UL 94 V-0	UL 94
Specific Gravity	2.95 ± 0.10 g/cm³	ASTM D792
Standard Thicknesses	0.5 mm, 1.0 mm, 1.5 mm, 2.0 mm, 3.0 mm	ASTM D374
Continuous Operating Temp	-50°C to +200°C	IEC 60216

InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

PART / GRADE CODE	MATERIAL SPECIFICATION & CONSTRUCTION	PROTOTYPE / R&D TIER	VOLUME / OEM TIER
`Core-300 (100 cm² Ref)`	Custom CNC Module Pad (100 × 100 mm = 100 cm² @ ₹0.80/cm²)	**₹80.00 Proto | ₹68.00 Pilot**	**₹61.20 OEM Blanket (500+)**
`Core-BMS-50x50`	Standard Pre-Cut 50 × 50 mm (25 cm²) MOSFET / BMS Conduction Pad	**₹20.00 Proto | ₹18.50 Pilot**	**₹17.00 OEM Blanket (500+)**
`Core-350D (>14 kV/mm)`	High-Dielectric Heavy E-Glass Anti-Puncture Grade (100 cm² ref.)	**₹88.00 Proto | ₹74.80 Pilot**	**₹67.32 OEM Blanket (500+)**

4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- EV traction battery module bottom cell-to-liquid-cold-plate thermal draining interfaces during ARAI AIS-156 rapid prototyping.
- Battery Management System (BMS) shunt resistors, D2PAK MOSFET arrays, and onboard charger (OBC) baseplate conduction.
- Industrial power supplies, telecom rectifiers, and driver chassis thermal bridging requiring mechanical tear resistance.
- Pre-packaged R&D Benchtop Sampler Boxes (₹9,800) enabling engineers to test 1.0, 1.5, 2.0, and 3.0 mm gap deflection on Day 1.

5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. Liner Removal Sequence: Peel the thinner 0.050 mm top extended-tab PET liner first, position the naturally tacky face onto the cold plate using alignment bosses, and smoothly peel away the stiffer 0.075 mm differential bottom carrier at a 180° low angle. 2. Recommended Compression: Design assembly stackup for 15% to 30% nominal thickness deflection (15–35 psi) to minimize contact resistance. 3. Commercial Policy: Custom Core™ prototype runs ship under prepaid R&D kit or milestone PO terms with ₹0 tooling NRE.

6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

Thermal & Electrical Metrology	Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification.
Flammability & Fire Safety	UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing.
Outgassing & Environmental	RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades.
Domestic Value Addition (DVA)	100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban.

RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
Official Orders, CAD & RFQ Desk: manav.sb9719@gmail.com | **Engineering Helpline:** +91 78148 XXXXX
Manufacturing Wing: InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India